IEEE 802 Series and Its Variants

Computer Networks • Student Study Notes

Learning objective: Understand what the IEEE 802 family is, which networking areas its major standards cover, and how common variants differ.

1. What is the IEEE 802 family?

IEEE 802 is a family of networking standards developed by the IEEE 802 LAN/MAN Standards Committee. It primarily specifies technologies for local and metropolitan networks, including important parts of the **Physical layer (PHY)** and the **Data Link layer**—especially the Media Access Control (MAC) sublayer. It helps equipment from different manufacturers communicate using agreed technical rules.

2. Major IEEE 802 standards at a glance

Standard	Main area	What it is used for / examples
802.1	Bridging, LAN architecture and management	Bridges, VLANs (802.1Q), port-based network access control (802.1X), link aggregation (802.1AX), MAC security (802.1AE).
802.2	Logical Link Control (LLC)	Historically defined the LLC sublayer for the data-link layer; now largely legacy in common networking practice.
802.3	Ethernet (wired LAN)	Wired Ethernet over copper or fiber. Common variants include 10BASE-T, 100BASE-TX (Fast Ethernet), 1000BASE-T (Gigabit Ethernet), 10GBASE-T and higher-speed Ethernet.
802.11	Wireless LAN (WLAN)	Wi-Fi. Common generations: 802.11b/g (2.4 GHz era), 802.11n (Wi-Fi 4), 802.11ac (Wi-Fi 5), 802.11ax (Wi-Fi 6/6E), and 802.11be (Wi-Fi 7). Bands and features depend on the amendment and local regulation.
802.15	Wireless personal / specialty networks	Short-range wireless standards. IEEE 802.15.4 is used by technologies such as Zigbee and Thread; it is designed for low-power, low-rate devices. Bluetooth's historical relationship to 802.15.1 should not be confused with current Bluetooth specifications being maintained by the Bluetooth SIG.
802.16	Broadband wireless access	The WiMAX family; designed for broadband wireless access, especially metropolitan-area deployments. It is less prominent in consumer networks today.
802.18	Radio regulatory technical advisory group	Coordinates and advises on regulatory matters affecting wireless standards; it is not itself a Wi-Fi-style access technology.
802.19	Wireless coexistence	Addresses coexistence between wireless standards and systems that may share spectrum.
802.21	Media-independent handover	Frameworks intended to help devices manage handover between different network access technologies.
802.22	Wireless Regional Area Networks (WRAN)	Designed for regional broadband access using cognitive-radio techniques and available spectrum, including TV white spaces where permitted.

3. Common variants students should remember

Ethernet — IEEE 802.3

10BASE-T = 10 Mb/s; 100BASE-TX = 100 Mb/s; 1000BASE-T = 1 Gb/s; 10GBASE-T = 10 Gb/s. The names identify Ethernet physical-layer implementations and their media/speeds.

Wi-Fi — IEEE 802.11

802.11n = Wi-Fi 4; 802.11ac = Wi-Fi 5; 802.11ax = Wi-Fi 6 (and Wi-Fi 6E when extended into 6 GHz); 802.11be = Wi-Fi 7. Wi-Fi generation labels are consumer-friendly names, while IEEE numbers identify standards/amendments.

Low-power wireless — IEEE 802.15.4

Commonly used as a radio/link foundation by higher-level networking technologies such as Zigbee and Thread. It is suitable for sensor and smart-home devices where low power matters.

4. Where do these standards fit in the OSI model?

Most IEEE 802 access standards focus on the **Physical layer (Layer 1)** and the **Data Link layer (Layer 2)**. The Data Link layer is often described as having two sublayers: **LLC** (Logical Link Control) and **MAC** (Media Access Control). The exact division and terminology depend on the standard. IP, TCP, HTTP and other higher-layer protocols operate above these link technologies.

5. Quick revision questions

1. Which IEEE 802 standard is associated with wired Ethernet?

2. Which standard family defines Wi-Fi?

3. What is the main purpose of IEEE 802.1Q?

4. Name the Wi-Fi generations associated with 802.11ac and 802.11ax.

5. Why is IEEE 802.15.4 useful for battery-powered sensors?

Answers: 1) 802.3. 2) 802.11. 3) VLAN tagging/bridging. 4) Wi-Fi 5 and Wi-Fi 6. 5) It supports low-power, low-rate wireless communication.

Note: IEEE 802 standards evolve over time. This handout summarizes commonly taught standards and examples; consult the official IEEE 802 pages for the latest status and exact scope of each standard.

References: IEEE 802 LAN/MAN Standards Committee (ieee802.org); IEEE 802.11 Working Group (ieee802.org/11); IEEE Technology Navigator, IEEE 802 and 802.3/802.11 standards overview.